

Review

Digital Technologies in the Sustainable Design and Development of Textiles and Clothing—A Literature Review

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Abstract: This paper examines the digital transformation of the textile and fashion industry, focusing on the alignment with sustainability principles through the integration of Industry 4.0 technologies. The introduction highlights the urgency of transitioning from conventional production methods to innovative, digitally enabled systems that promote a circular economy and resource efficiency. The main research questions address the contribution of Industry 4.0 elements to sustainable solutions, the directions of digitalization within the apparel sector, and the significant impact of digital technologies on the achievement of sustainability goals. The theoretical framework examines sustainability in the textile industry and emphasizes the need for a green transformation facilitated by digital technologies to reduce environmental impacts. Industry 4.0 concepts, as discussed in The Concept of Industry 4.0 in the Textile and Apparel Sector, are revolutionizing production through technologies such as IoT, AI, and blockchain, enabling traceability, customization, and energy-efficient operations. The paper also explores the evolution of the fashion and apparel industry into a high-tech sector, highlighting advances such as CAD-CAM systems, digital printing, and 3D technologies that improve precision, reduce waste, and support sustainable practices. In its conclusion, the paper emphasizes the crucial role of interdisciplinary collaboration, regulatory frameworks, and investment in skills development to overcome the challenges of implementing digital and sustainable practices. It posits that a strategic embrace of digital ecosystems and Industry 4.0 technologies is essential for creating a resilient and sustainable textile industry that is aligned with environmental and societal goals.



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1. Introduction

Digital technologies have played and continue to play a key role in aligning the development of the textile industry with the concepts of sustainable production, circular economy, renewable materials, and renewable energy sources. Digital technologies enable the development of online platforms for the presentation of design and the sale of products, resulting in a faster turnaround time through the supply chain. In addition, the introduction of new digital technologies and the development of the concept of sustainability are rapidly paving the way for innovations in the textile and fashion industry, such as innovative systems for 3D visualization, tools for virtual presentations and new, digitally enhanced production systems. All this together brings economic benefits and helps to create a more positive public attitude towards textile and fashion companies. The textile industry therefore no longer has the option of using conventional technologies and processes for production [1,2].

It must introduce agile, modern, and sophisticated processes in every respect. The textile industry, one of the oldest and most traditional industries, plays an important role in society and the environment and has an impact on various aspects of modern life. It creates jobs, influences culture, and provides access to clothing, but also faces challenges related to sustainability. As Banerjee and Kumar (2024) point out in his recent study, key issues include extended production times, disjointed supply chains, delays in market entry, and an increasing number of fashion seasons characterized by greater volumes, variety, and speed. The sustainable transformation of the textile industry is a real systemic challenge that requires determination to change and massive investment from all stakeholders over a long period of time. The transformed operating model is not only conceivable but also practical to implement and brings multiple benefits to people, institutions, and the economy [3]. In today's textile industry, which is evolving into a high-tech sector, it is crucial to formulate a strategy for the development of new products and process innovations that take advantage of new technologies [4,5].

It can be therefore said that the prospects for the textile and fashion industry are closely linked to the dual transformation of digitalization and sustainability. By integrating digital tools and sustainable practices, companies not only minimize their environmental impact, but also increase their efficiency and promote innovation. Digital transformation in the textile industry involves the integration of advanced technologies such as IoT, AI, blockchain, and big data analytics into traditional textile manufacturing and supply chain processes. Digital technologies enable more sustainable practices in the textile industry. For example, IoT sensors can monitor and optimize energy consumption, while blockchain can ensure the traceability of materials, promote ethical sourcing, and reduce waste. However, the concept of integrating these changes is very complex, and their application in the textile and apparel industries is particularly complicated and sometimes operationally unsolvable [6]. Therefore, there is a gap between the proposal and the implementation of these practices, the technologies, and the policies in the manufacturing sector. For example, the EU has developed an exemplary policy for the circular economy in the textile value chain which sets ambitious targets for reducing emissions, increasing resource efficiency and promoting recycling and reuse. Specific regulations for the textile sector have also been introduced, such as restrictions on hazardous substances and extended producer responsibility requirements. However, despite the current state of the art, full sustainability has not yet been achieved and one of the main reasons is still the excessive volume of production and consumption [7,8].

In this paper, theoretical research based on the literature was conducted to analyze the basic development directions of the fashion and apparel industries and its transformation into a high-tech sector in more depth and detail.

The main research questions underlying this analysis were the following:

- First, which elements of the concept of Industry 4.0 contribute to the implementation of sustainable solutions in the fashion industry (textile and apparel industry)?
- Second, what are the main directions of digitalization in the apparel and textile industries, focusing on areas close to the author's expertise, namely the clothing industry and digital textile printing?
- Thirdly, which digital technologies have a significant impact on achieving sustainability and circular economic concepts, especially in the areas mentioned, and how do they do so?

Based on the research questions posed, the basic research hypotheses for this review are as follows:

H1: *Digitalization is the key enabler of a sustainable and efficient transformation in the textile industry, significantly improving production processes, minimizing environmental impact, and enhancing supply chain transparency, while its full potential is hindered by infrastructural, financial, and skill-related challenges.*

H2: *The adoption of Industry 4.0 technologies in the textile industry accelerates digital transformation, enhances sustainability, and optimizes production processes, yet full implementation is hindered by economic, infrastructural, and regulatory challenges.*

H3: *The successful transition to a sustainable textile industry requires collaborative efforts between industry, policymakers, and research institutions to overcome technological and economic barriers.*

Based on the following chapters and the topic researched through an extensive literature review, the validation of these hypotheses is presented in Section 5: Hypothesis testing.

The overall aim of this analysis was not to critically review the existing literature, but to summarize data found in the existing literature in order to provide an overview of the latest development trends in the context of digitalization and sustainability for specific sectors. The contribution of this analysis is in the flow of information that is crucial for understanding the current directions of development and for understanding certain challenges and issues in their practical implementation between the research, educational, and economic sectors. The article also consists of a more detailed overview of the field of digital printing in the textile and fashion industries, which is of great importance for the modernization of textile production and the achievement of more efficient and sustainable production processes and is relatively underrepresented in articles dealing with similar analysis and overviews. The remaining article is organized as follows: Section 2 presents the theoretical framework that forms the basis of the review analysis. Section 3 explores the integration of sustainability and digitalization in the textile industry, detailing the literature review methodology, theoretical foundations of sustainability and Industry 4.0, and the role of digital technologies in textile production. Section 4 defines the concepts of sustainable fashion and apparel. Section 5 focuses on hypothesis testing, while Section 6 presents the conclusion.

2. The Theoretical Framework of the Analysis

2.1. Aspects of Sustainability in the Fashion and Textile Industries

Awareness of sustainable business practices is becoming central to companies' strategies. Many sectors now incorporate environmental, social, and management factors to achieve sustainable production systems. Companies that invest in green transformations can adapt and remain competitive. Digital technologies play a key role in enabling efficient production, reducing energy use, and minimizing environmental impact. However, digital innovations have focused more on user applications than on large-scale industrial solutions. Rapidly changing consumer preferences have driven hyper-consumption and hyper-production, exacerbating climate change, biodiversity loss, and textile waste. The concept of sustainable development offers a solution, but its implementation remains slow [9]. The European Parliament's 2024 report highlights the textile industry's significant environmental impact: it accounts for 20% of global water pollution, 10% of CO₂ emissions, and produces 1.2 billion tons of greenhouse gases annually. Less than half of old clothes are collected for reuse, and only 1% are recycled into new garments due to underdeveloped recycling technologies. The industry's impact includes excessive fossil fuel use, water pollution, and inefficient land use [10]. Sustainable development, emphasizing eco-efficiency, recycling, and renewable resources, provides a framework for addressing these issues. Transformative change requires collaboration among authorities, companies, and citizens, with education and media playing crucial roles in fostering environmental awareness. Even if sustainability does not yet dominate the current landscape, it is undoubtedly the key to the future of fashion [11,12].

2.2. Innovation, Research, and Development

A modern and competitive textile industry relies on the timely adoption of innovative technical and technological solutions, efficient tools, and organizational advancements

rooted in ICT and strategic work development. Since the advent of automated machines and ICT integration, the textile and garment sectors have sought to enhance labor efficiency and production productivity. These efforts have driven the development of ICT-based strategies, such as CAD/CAM systems, coupled with increasing automation in technical processes. However, the industry's adoption of digital tools has been slower compared to other sectors, primarily due to the unique challenges posed by garment manufacturing. Small batch sizes, frequent fashion changes, and the inherent complexity of handling flexible textile materials make automation time-consuming and intricate [1,13,14]. The wide variety of garment types—around 350 with numerous variations—complicates the uniform implementation of CAD/CAM technologies across different production stages. These challenges underscore the need for tailored solutions that account for variability at each stage. To modernize the textile and clothing industries effectively, collaboration between research institutions and industrial stakeholders is crucial. This partnership can drive digital transformation and establish long-term development programs. However, scientific research alone is insufficient; a strategic shift in innovation policy is essential to foster diversification and develop innovative solutions. Such advancements can elevate the industry by creating globally competitive, high-quality products while addressing sustainability challenges. By adopting these strategies, the textile and garment industries can transition toward a high-tech, sustainable future [14,15].

2.3. Digital Transformation of the Textile and Fashion Sector

The textile industry is undergoing a profound digital transformation fueled by technological advancements and evolving consumer demands. Tools such as automation, artificial intelligence (AI), the Internet of Things (IoT), and 3D printing are revolutionizing how textiles are designed, produced, and utilized. These innovations promise greater efficiency, sustainability, and adaptability but also present challenges that must be addressed. One of the key benefits of digital transformation is the automation of labor-intensive tasks. AI and machine learning (ML) are pivotal in optimizing production processes by analyzing extensive datasets, identifying patterns, and predicting potential issues. For instance, AI can forecast defects, recommend machine adjustments, or estimate product demand, ensuring smoother operations. Digitalization further enables product customization and the on-demand production of personalized textiles, aligning with consumer preferences. Moreover, digital technologies contribute to environmental sustainability by addressing high water and energy consumption. IoT devices enhance supply chain transparency and operational efficiency, with sensors providing real-time monitoring of production conditions. Blockchain technology complements these efforts by documenting every step of a product's lifecycle, from raw material sourcing to final manufacturing, promoting traceability and ethical practices. Despite these advancements, the industry faces significant obstacles. High initial costs for acquiring modern machinery and software are a major barrier. Integrating new technologies into existing production systems is often complex and time-intensive, requiring robust infrastructure and specialized expertise. Resistance to change, particularly in traditional markets, is another challenge, as digital transformation demands alterations to business processes and employee roles. Workforce retraining becomes essential, and concerns about job displacement due to automation must be addressed. The path forward for the textile and fashion industries lies in embracing continuous innovation and digital technology adoption. These advancements hold the potential to improve efficiency, reduce environmental impact, and cater to the demand for sustainable, customized products. Overcoming challenges like cost, integration complexity, and workforce adaptation is critical for the industry to fully harness these benefits. With strategic investments and collaborative efforts, the textile sector can transition into a competitive, sustainable, and technologically advanced future [16–20].

3. Integrating Sustainability and Digitalization in the Textile Industry

3.1. Literature Review Methodology

This section presents the methodological framework used for synthesizing insights into sustainability and digitalization in the textile and fashion industries. The literature review explores trends, challenges, and opportunities in sustainable fashion and production, focusing on advanced technologies such as CAD/CAM systems, artificial intelligence (AI), augmented reality (AR), virtual reality (VR), the Internet of Things (IoT), and RFID technology. Additionally, it examines digital textile printing in sustainable production, incorporating advancements in 3D simulation, visualization, and 3D printing. The analysis targets the literature published between 2019 and 2024.

Defining keywords posed challenges due to the interdisciplinary nature of the topic. The selected keywords included “sustainable fashion”, “sustainable textiles”, “CAD/CAM in fashion”, “digitalization of the textile industry”, “IoT in the fashion and textile industry”, “AI in the fashion and textile industry”, “VR and AR in the fashion and textile industry”, and “circular economy in the textile industry”.

A structured and iterative search was conducted using databases such as Web of Science, Scopus, Google Scholar, EbscoHost, Emerald Insight, and Taylor & Francis. The review included scientific articles, conference proceedings, and book chapters, applying inclusion criteria that restricted the selection to English-language publications with full-text availability.

An initial pool of 387 references underwent coding and thematic evaluation. Titles and abstracts were reviewed to eliminate duplicates and irrelevant works. This process refined the sample to 135 publications, ensuring alignment with the study’s objectives. A PRISMA flowchart detailing the selection process is provided in Figure 1.

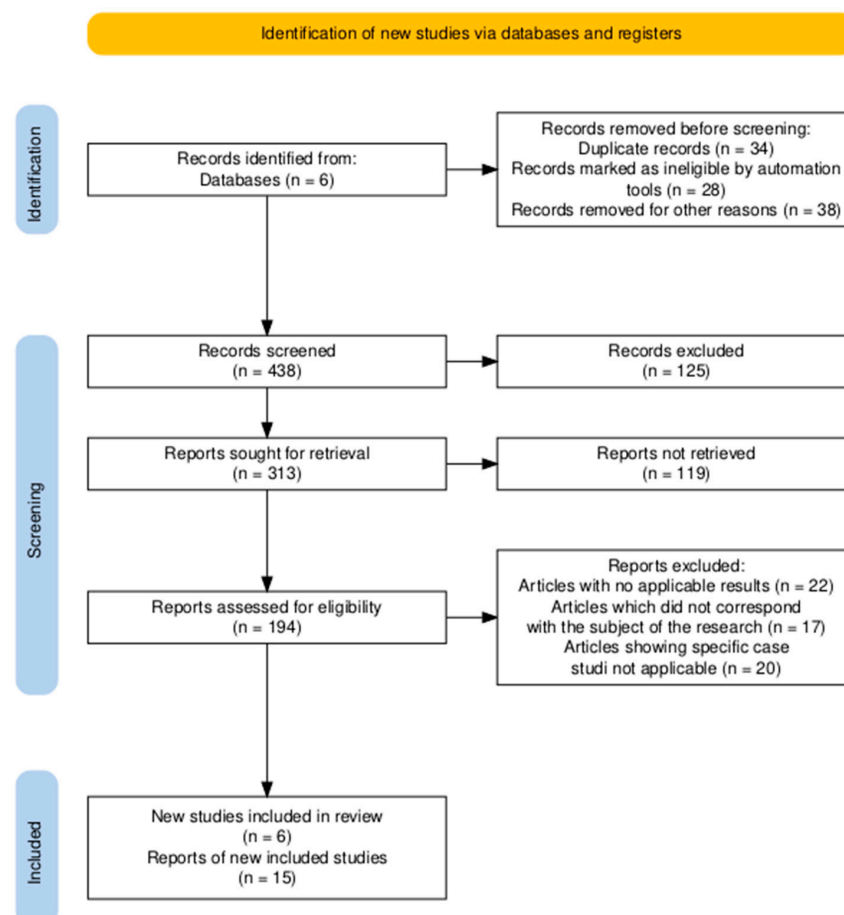


Figure 1. PRISMA flowchart.

3.2. Theoretical Foundations: Sustainability and Industry 4.0

3.2.1. Principles of Sustainable Fashion

The digitalization of production processes has led to a significant shift in the traditional industry towards new, sustainable business models. The current production model has a high negative impact on the environment and also on society. The move towards “smart production” and the use of innovative technologies are leading to greater sustainability and efficiency.

A critical aspect of sustainable fashion lies in embracing a circular economy. This approach emphasizes the reuse and recycling of materials, reducing waste, and prioritizing renewable resources. Additionally, local production models enabled by digital technologies allow for greater interaction among stakeholders, fostering knowledge sharing and reducing the carbon footprint associated with global supply chains.

Sustainability also extends to the design and production phases. Digital platforms facilitate real-time collaboration, shortening lead times and enabling personalization. These tools align with the broader goals of renewable energy use, waste minimization, and material optimization.

The adoption of advanced tools like AI and IoT further enhances traceability and transparency. Blockchain systems integrated into IoT frameworks provide detailed insights into product lifecycles, ensuring ethical sourcing and quality control. Technologies like RFID streamline supply chain monitoring, enabling real-time oversight and efficient resource allocation.

Due to the increasing innovation and efficiency resulting from digital hardware and software, fundamental technologies and business models are being completely overturned throughout the textile industry [20–22]. In the textile and apparel industries, the application of these technological building blocks has meant revolutionary changes in terms of sustainability and energy and production efficiency, affecting virtual fabrics and fashion design, digital printing, electronic machine control, and e-commerce.

3.2.2. Industry 4.0 in the Textile Sector

The textile sector has emerged as a pivotal area for implementing Industry 4.0 principles. Characterized by the integration of digital technologies, this revolution transforms traditional manufacturing into smart, interconnected systems.

Bertola and Teunissen (2018) carry out a detailed analysis of the impact of the Industry 4.0 concept on the textile and fashion industries [23]. In their study, through a comprehensive analysis of the literature and examples of best practices, they point out the compatibility of the Industry 4.0 paradigm in the context of the fashion industry, especially due to its characteristic mass production for which the fashion industry is known worldwide. Traditional business models in the fashion industry were based on design, for which Industry 4.0 technologies can create intelligent networks. Digital technologies enable interaction between employees via platforms so that different fashion producers can easily share knowledge, which is considered a business model for local production. In addition, the transition to innovative, smart manufacturing is changing the textile industry system from a market-oriented to a ready-to-wear model, which also increases the interaction between companies and consumers. For sectors with massive production, such as the textile industry, the transition to a circular economy is extremely important. Incorporating the concept of Industry 4.0 into the textile industry implied a readiness for a rapid shift from primarily analog to digital technology and virtual trends. Other Industry 4.0 factors have significantly changed the traditional textile industry: remote factory monitoring and maintenance, automated supply chains, virtual trade fairs and product showrooms, direct-to-consumer business model, big data utilization and machine learning, AR/VR, use of IoT (Internet of Things) in factories, additive manufacturing, and cybersecurity. Machine vision, a key technology within the Internet of Things (IoT), plays a pivotal role in these

advancements, especially when integrated with deep learning approaches. Machine vision can encompass various techniques, not all of which require deep learning. For example, traditional computer vision techniques like edge detection, template matching, and feature extraction may suffice in simpler applications. However, modern advancements in machine vision often involve deep learning, especially for complex tasks like object detection, image segmentation, and localization. Digital technologies have played and continue to play a key role in aligning the development of the textile industry with the concepts of sustainable production, circular economy, renewable materials, and renewable energy sources. Digital technologies enable the development of online platforms for the presentation of design and the sale of products, resulting in faster turnaround times through the supply chain. Among these technologies, machine vision, particularly when combined with deep learning models, enhances automation and efficiency by enabling precise detection, tracking, and classification of objects within manufacturing processes. Machine vision, as a subset of IoT, enables devices to “see” and interpret their surroundings, facilitating smart decision-making. By incorporating depth information through RGB-D imaging, machine vision systems provide enhanced spatial understanding, which is critical for tasks requiring high precision. For instance, identifying defects in fabrics or optimizing cutting patterns based on material properties can minimize waste, a significant step towards sustainable practices. Deep learning technologies, particularly convolutional neural networks (CNNs) like YOLOX, have revolutionized object detection and localization tasks. Incorporating such advanced frameworks into the textile industry enables the real-time monitoring and prediction of production anomalies, such as defect identification in fabrics or optimizing dye application. These innovations reduce material wastage and improve production efficiency. Additionally, the integration of IoT sensors and machine learning models facilitates predictive maintenance, reducing downtimes and ensuring optimal use of resources. Blockchain technologies further enhance this ecosystem by enabling traceability, promoting ethical sourcing, and ensuring compliance with sustainability regulations [23–25].

In their study on certain aspects of the cornerstones of the Industry 4.0 concept, Faridi et al. (2020) identify traceability and reliability as two key problem areas in textile production [26]. According to Bosnoa and Gebresenbet (2013), traceability is also the key to quality in the textile industry, and technologies such as the IoT (Internet of Things) are fundamental to the traceability management system [27]. The blockchain system integrated into the IoT provides information about the entire product lifecycle and communication between different stages of the production process. They also provide insight into defects and omissions during individual stages of the production process, which enables their early detection and timely elimination, all of which serves to improve the quality of the final product.

The digitization of information through IoT-enabled applications as well as related technologies such as radio frequency identification (RFID) have revolutionized the textile industry, enabling high-quality and efficient supply chain monitoring, real-time control and inspections, as well as the monitoring and management of each stage of production [28]. In their systematic review, Panarello et al. (2018) conduct a comprehensive analysis of current trends in the use of blockchain applications and technologies in the IoT context. By examining different application areas, two main application categories are identified and initialized: device management and data management [29]. Manglani et al. (2019) also publish a comprehensive study on the application of the IoT concept in the textile industry in which numerous case studies are presented and innovative approaches in various processes (yarn, fabric, clothing, finishing) are analyzed [30].

Kemper et al. point out certain challenges in the implementation of Industry 4.0 in the field of textile machinery manufacturing and textile production resulting from the incoherence

of certain sectors [30,31]. A study on the implementation of a digital environment based on augmented reality in a weaving mill, based on the interaction between man and machine, has shown that the digitalization of production, on the contrary, increases the ability of the workforce and serves as a support by eliminating monotonous and repetitive tasks, making work easier [32].

The application of a multi-objective input algorithm in the optimization of warp tension, energy consumption, and fabric quality enabled the construction of the weaving machine as a cyber-physical machine (CPS). Also, the application of an intelligent bobbin with an integrated radio frequency identifier (RFID) enabled the tracking and reading of inhomogeneities, various irregularities, the position of a particular fault, and the exact time of use of the material in the bobbin [31,33].

Clothing must be adapted according to the season, intended use, and consumer demand, and the production of large quantities is particularly labor-intensive. Robotic sewing systems have already been developed for special applications. Textile-based wearable robots have been developed as part of the research being conducted in areas such as robotics, biomedical engineering, and human–computer interaction. From the compilation of this research, several classes of wearable robots are emerging, including locomotion assistance, grasping and reaching, and shape-shifting for dressing. The use of AI technologies offers the potential to increase sales or optimize processes. The term industrial artificial intelligence is often used to describe AI-based industrial applications, technologies, and functions. Pattern inspection, defect detection, and color matching are some of the most commonly used AI applications in textile manufacturing. Other important developing segments of IoT applications in the textile industry from a process perspective are automated sewing machines and fabric defect detection systems. These solutions currently support operators, although they have the greatest potential for the application of machine learning and artificial intelligence [34–38].

Xu et al. (2024) published a detailed study on the concept of zero-defect manufacturing (ZDM) in which they examine the basic ZDM strategies: detection, repair, prediction, and prevention. They conducted an assessment of the current state of the art and identified barriers and shortcomings to its practical application. Some of the findings are limitations in automatic fault detection, incomplete repair strategies based on AI, limited prediction research, and focused prevention mechanisms [39]. Considering the aforementioned limitations, especially in terms of an AI-based repair strategy, they propose the application of machine learning and computer vision techniques, where AI can quickly identify the location and extent of fabric defects through data analysis and prediction and provide more effective repair solutions that reduce textile waste and promote the sustainable development of the textile industry. The ZDM concept has the potential to transform the textile industry and steer development towards a more efficient, sustainable, and reliable production paradigm. Of course, realizing such an approach and concept requires a comprehensive understanding of the unique challenges and opportunities that this area presents, as well as the development of innovative and adaptable solutions that can effectively address them [39–43].

3.3. Digital Technologies in Sustainable Textile Production

Industrial digitalization shapes and transforms not only its own domain but also broader economic, social, and cultural spheres. Technological innovation and human capital drive these changes, with “digital transformation” being a key term in the fashion industry, known for its profound cultural impact. The rise in digital technologies has digitized information and processes, reshaping value creation and enabling new products and services. This transformation involves overcoming structural challenges and adapting

to rapidly evolving customer expectations. The demand for innovations like 3D-printed clothing and smart factories has surged, driving the adoption of technologies such as virtual reality (VR), augmented reality (AR), and mixed reality (MR), ushering in a new technological era [44–47].

3.3.1. CAD/CAM—Paradigm of the Digitized Fashion, Textile, and Clothing Industries

One of the milestones in the emerging digitized high-tech textile industry was the development of the application of computer technology, and one of the prominent technologies that directly contributed to the transformation of design and production in the late 80s and early 90s of the 20th century was Computer-Aided Design and Computer-Aided Manufacturing (CAD/CAM). These technologies found application in many industries, including manufacturing, automotive, aerospace, medical, and many others that previously did not have the ability to apply common, one-size-fits-all systems. It can be said that the introduction of CAD-CAM technologies has irrevocably changed the industrial landscape [48,49]. CAD and CAM are two systems that allow designers and engineers to digitize the design of the product, which is then transformed into a physical form by automated equipment. One of the features of the CAD-CAM system is the ability to create not only 2D but also 3D models, which allows for greater speed and precision in the product development process. Due to its structure, CAD allows for a number of product modifications, simulations, and visualizations, achieving a higher level of control before the actual production process begins (Figure 2).



Figure 2. CAD for fashion design, pattern development, and 3D visualization.

By using CAD technologies, designers can explore different design options, identify potential problems, and make timely corrections and changes without the expense of producing physical test samples. With the ability to create virtual models of products, designers can more easily, quickly, and efficiently develop the design of complex products and devices, verify their performance, and make certain changes and modifications. Iterative testing of the virtual model reduces the risk of errors and unnecessary costs associated with physical prototyping and speeds up the production process. The CAM system, in turn, enables the automation of the production process, which reduces dependence on human labor and increases productivity. The production process is faster, with greater accuracy, high manufacturing precision and better product consistency [50–52]. In the apparel industry, the CAD system programs originally developed were focused on pattern

digitization, computer grading, and marker making in garment production. Today, with the advancement of computer configurations and programs, modern CAD systems enable the complete computer development of patterns, from basic pattern construction or digitizing, pattern modeling and development, preparation for pattern making according to industry standards and various user requirements, to computer creation and the development and analysis of 3D garment models. Accordingly, most of the processes in fashion manufacturing, especially in patterning and design development, have been computerized. In addition to the product design and development phases, computerization has also penetrated the areas of management, marketing, and sales. The creation of a digital image allows designers to create a more detailed and accurate representation of the original design, making the next stages of development easier. Computer-generated design sketches are more reliable, minimizing the possibility of errors and misinterpretations in the garment construction phase. This leads to significant time and material savings, which in turn leads to a more sustainable and environmentally friendly production system. Jhanji (2018) conducts a case study in which he compares the time taken to perform various pattern-making activities using a CAD/CAM system with the time taken for manual operations. He conducts this research to explain the practical benefits in the apparel industry by switching from manual methods to CAD/CAM technology. The analysis has shown that in a new pattern creating activity, expressed in minutes as units of time, with the manual method for a 100 min job, CAD/CAM takes 15.22 min. For grading (per size), 5.67 min are required for an order of 120 min with the manual method and 5.92 min with CAD/CAM for pattern planning for an order of 100 min with the manual method. It was also found that the proportion of waste could be reduced by 10% when cutting with the CAD/CAM systems compared to the manual method. The unit with CAD/CAM systems had a production of 15,856 pieces/day while the unit with manual system had a much lower production of 4013 pieces/day. The lead time was also higher for the unit with the manual method (55 days) than for the unit with CAD/CAM systems (39 days) [53–55].

Among the major global manufacturers of specialized two-dimensional (2D) and three-dimensional (3D) CAD systems for the needs of the fashion and apparel industries are Lectra (V2024.2.), Style3D (V7.2), Assyst (20.24), Optitex (0/24), Browzwear (2024.2), Audaces 360, and Investronica (V11). Among the renowned software that cover all major operations in the apparel industry are Clo3D (2024.2.284), Style 3D (20.24), Optitex 2D/3D Pattern Design Software (O/24), VStitcher (2024.2), 3D Vidya (20.22), 3D Design for Illustrator (4.2), and Modaris 3D (V2024.2.).

Wonderweaves, Reach technologies, Summagraphics, and Prostyle are some of the companies that offer textile design software for designing woven and knitted textiles. TUKA studio is an example of a CAD system that is suitable for both textile printing and weaving and enables complex weaving and color schemes [56,57]. An interesting study was conducted by Chae (2022) in the field of application of CAD systems in modeling color predictions for the weaving process when using multicolored yarn. The color of the woven fabric is determined by the different colors of the yarn, which are interwoven in a complex weave and in different proportions. Thus, predicting or reproducing the color of woven textiles from multicolored yarns is a difficult task that requires appropriate color mixing recipes. In his study, Chae (2022) proposes three-dimensional spectral reflectance-based color prediction models for single- and multi-layer woven fabrics, which have been shown to be beneficial for producing various woven fabrics in desired colors, and their potential application to CAD systems [58]. Digital Product Creation—DPC—is another concept related to fashion product development and prototyping that has recently been the focus of research and investment programs in the fashion industry [59]. It not only refers to the replacement of the physical sampling of new models in the development area but is

already having a much greater impact on redefining the fashion industry. The creation of digital representations of fashion products and the creation of digital assets is influencing creative and commercial decisions, and there is almost no stage of the value chain where the importance of a digital fashion product is not recognized. With new approaches in design, as well as 3D simulation, virtual model fitting, and product visualization aimed at e-commerce, the creation of digital fashion products is expected to become even more important in the coming years and is crucial to achieving the strategic goals of fashion brands. DPC is a process that uses advanced technologies to conceptualize, design, develop, and manufacture 3D products in a virtual, collaborative environment. It offers an environmentally friendly and cost-effective alternative through material digitization, 3D pattern simulation, and 3D assets. The parallel use of a program to develop 2D pattern pieces and a program to perform a 3D simulation of a digital garment model enables the simulation of the assembly of the 2D pattern pieces and a realistic representation of the virtual garment on a 3D computer model of the human body with specific anthropometric features (Figure 3) [60–62].

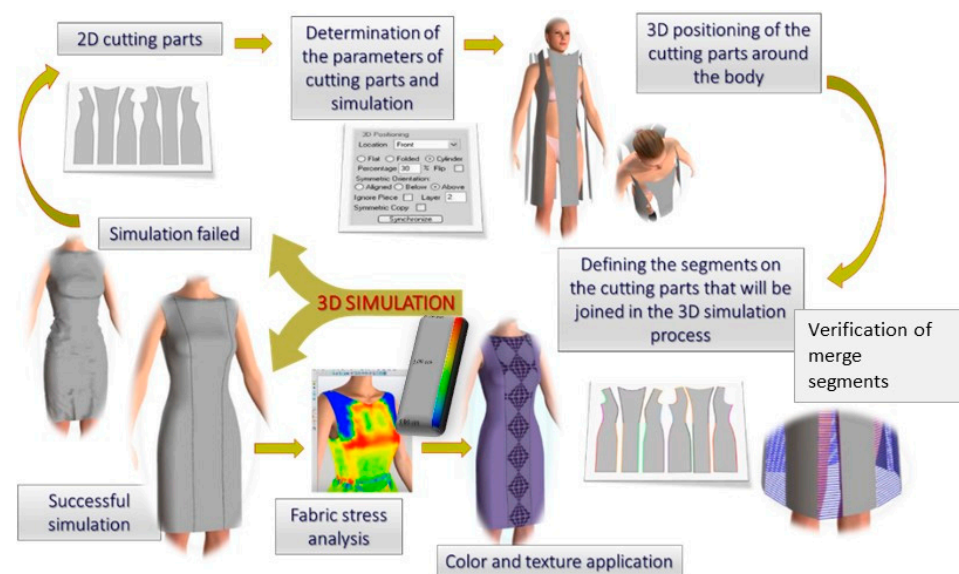


Figure 3. Process cycle of the digital clothing development.

3.3.2. The Future Prospect of CAD Technologies in a Frame of AI, AR, and 3D

Computer-Aided Design (CAD) is advancing through its integration with artificial intelligence (AI), augmented reality (AR), and 3D printing, reshaping industries like fashion with innovative, efficient, and sustainable solutions. CAD's core functions, such as design modeling, analysis, and documentation, are becoming more powerful when combined with these technologies. AI enhances CAD by enabling real-time decision-making through quantitative reasoning, simulations, and deep modeling. This technology supports tasks like evaluating production line components, improving workplace safety, and reducing costs. AI also facilitates faster product development by identifying design improvements and optimizing workflows, offering significant potential for sustainable innovation. AR and VR enable interactive and immersive design experiences, allowing designers to simulate and analyze objects in a virtual 3D environment. These technologies provide enhanced visualization, enabling more accurate and engaging presentations of designs. When integrated with CAD, AR/VR helps refine product development processes and supports virtual try-ons for consumers, significantly improving user experience and reducing unnecessary production [63,64].

Three-dimensional printing, a cornerstone of additive manufacturing, works seamlessly with CAD to produce complex, high-precision objects layer by layer. This technology addresses inefficiencies in traditional manufacturing by reducing lead times and costs, supporting mass customization, and enabling rapid prototyping. Without CAD's design capabilities, the potential of 3D printing would remain unrealized. Three-dimensional printing has become integral to fashion, producing flexible, functional garments and accessories. Designers use it to create complex structures, blending soft fibers with filaments for enhanced consumer acceptance. Applications include sportswear, medical textiles, and smart clothing equipped with sensors, showcasing its versatility and innovation [65–72].

Three-dimensional printing promotes sustainability by using recycled materials like polyamide (PA), thermoplastic polyurethane (TPU), and polylactic acid (PLA). These materials minimize waste while maintaining durability. However, achieving the unique flexibility and comfort of traditional textiles remains a challenge. Researchers are exploring new methods, such as adjusting print geometry and size or using Fused Deposition Modeling (FDM) to control deformation properties. Research also suggests printing with fibers to create textile structures. The fashion industry is experimenting with combinations of 3D-printed geometries on textiles that give these textiles additional functionality for various applications. These self-forming structures printed on textiles are attracting attention due to their significant functionality and esthetic value [73–77].

Kim et al. (2019) lists and analyzes some real cases of fashion design using this technology in her study on the development and improvement of fashion products with an FDM-type 3D printer. Based on the analysis, three approaches to the application of this technology are identified: the production of entire garments exclusively with 3D printing, the partial use of 3D printed products for garments, and the separate production of accessories and other items. One of the first examples of the production of a complete garment using 3D printing technology is the “Bomber jacket” by German designer Danit Peleg. Another example cited is the 3D-printed dress ‘Loom’, developed by designer Maria Alejandra Mora-Sanchez and Cosine Additive and printed using the FFF (Fused Filament Fabrication) process using TPU. One of the best-known examples of the partial application of 3D printing on a garment is that of German designer Babette Sperling who used the environmentally friendly, biodegradable material WillowFlex to print a Braille text directly onto the fabric, confirming the possibility of strong adhesion of this filament to natural fabrics such as cotton or silk. An example of the production of accessories is a 3D printed pair of shoes produced by the shoe company Feetz using an FDM-type FFF 3D printer. The material is Flexknit™ polymer filament, which is recyclable [78–82].

Since the advent of CAD systems, it has been possible to see the product in three dimensions before it is manufactured. Nowadays, CAD modeling is used beyond 3D modeling. Despite all the possibilities of 3D modeling, CAD applications still do not enable immersive visualization and interactive simulations as they are possible in augmented reality (VR), augmented reality (AR), and mixed reality (MR). By combining these technologies, i.e., by loading CAD software into VR, AR, or MR platforms, it is possible to mimic the real 3D world to perform immersive simulations and detailed conceptualizations [83–85].

Beyond 3D modeling, CAD now incorporates AR/VR for immersive visualization. Designers can load CAD files into AR/VR platforms to simulate and refine products in real-world scenarios. Fashion retailers are gradually adopting VR technology to promote virtual fashion shows for store visitors and virtual product presentations. These tools are particularly valuable in fashion retail, where virtual fitting rooms allow customers to try on digital clothing tailored to their body measurements. Companies like Zugara, Nettelo, Atlatl, and 3DLook provide AR-enabled apps for mobile and retail use, enhancing interactivity and customer satisfaction. Virtual fitting rooms combine augmented reality (AR) or virtual

reality (VR) to enable precise measurements and personalized recommendations, streamlining production while reducing waste. These systems analyze consumer preferences and buying behaviors, offering retailers actionable insights for product customization and inventory management. This integration of big data analytics promotes efficiency across the supply chain. The combination of CAD, AR/VR, AI, and 3D printing is revolutionizing production methods, aligning them with sustainability goals. Technologies like AR/VR reduce the need for physical samples, while AI optimizes processes to minimize resource use. Three-dimensional printing reduces waste and supports circular design principles by incorporating recycled materials into production [62,86–89].

The integration of CAD with AI, AR/VR, and 3D printing is setting new standards for innovation and sustainability. By improving design precision, reducing costs, and enabling mass customization, these technologies are transforming industries, particularly fashion. With continued research into material properties and consumer-focused applications, CAD and its allied technologies will play a pivotal role in shaping the future of sustainable design and manufacturing.

3.3.3. Digital Printing

Digital printing is one of the leading technologies in the field of modern textile printing where the result of the final appearance of a particular product depends on the interaction of several factors focusing on the specific properties of the textile material, its three-dimensionality, the variety of surface structure, and the diversity of the print and color ratio depending on the type and surface of the specific textile material. In the field of textile design, the individual surface characteristics and the influence of the color of the substrate on the appearance of the product lend authenticity to a high-quality textile product. Digital textile printing is a multidisciplinary field and it is important that product developers work simultaneously to ensure effective communication throughout the supply chain. Inkjet digital textile printing (IJDTP) is an innovative, emerging technology that offers new product development opportunities for specialty markets. Digital textile printing technology links the design software and printer, printing environment, fabric pre-treatment, post-treatment, and operator. This printing process is also a new environmental technology and a green and environmentally friendly process of textile printing which was first used in the textile industry and has unique advantages such as low cost, high precision, and speed compared to traditional printing, which could only produce high-quality and personalized products. Digital printing requires no set-up costs for new print designs and allows fabrics to be printed instantly. Interest in IJDTP is driven by the need for new product innovation and rapid response to consumer demand. Digital inkjet printing has been developed to either replace or work alongside traditional printing in the textile industry. The print-on-demand aspect of digital printing makes it more flexible. Inkjet printing is a non-contact printing technology in which dyes are ejected either by dripping or by spraying through the nozzles of the print head onto the substrate to be printed, a process that enables the direct transfer of a design from a computer file to fabric without the need to manufacture screens or operate heavy machinery [90–92].

According to Tyler (2005), there are three areas in which the digital printing of textiles differs from other textile printing processes and creates new opportunities for printing: flexibility, process improvement, and innovation [93].

In terms of flexibility, the digital printing of textiles enables lower set-up costs, small batches, and design freedom. There are process improvements from digital printing as there is less waste and no need for stock. The potential for innovation is increased as digital textile printing allows for customization. IJDTP, in conjunction with CAD, enables the creation of print designs with hundreds of colors, photorealistic prints, large format print

designs, and technical designs that are either expensive or unattainable in traditional screen printing. IJDTP's flexibility and customization capabilities allow for mass customization, limited editions, or one-of-a-kind pieces, meeting the growing consumer demand for novelty and variety in apparel and textile designs. Due to these characteristics, IJDTP has the potential to complement and, in some cases, replace conventional screen printing in established markets and create entirely new markets for textile products. A great deal of time and capital has been invested in developing the chemistry, software, and hardware associated with IJDTP. Digital technology simplifies the production line for textile printing. Color separation and the production of printing plates are no longer necessary. Therefore, it becomes environmentally friendly and cost-efficient for the textile industry [94–96].

In today's textile printing industry, economic, social, environmental, and toxicological considerations have become more important than simply improving textile print quality to achieve certain (esthetic) goals. Advances in inkjet technology have revolutionized the textile printing industry by significantly reducing the time and labor required. The digital textile printing process is therefore environmentally friendly, generates hardly any waste, is much cleaner and has a good socio-economic impact compared to the traditional rotary screen printing process. However, these changes in production are only possible with high-quality, environmentally friendly inks that can be printed on all substrates and have excellent coloristic properties. With advances in software, hardware, and chemical technology, as well as active research by textile chemists to develop universal dyes/inks suitable for digital inkjet printing on various textile materials, digital textile printing will realize its full potential and take the central role in textile printing. Pigmented water-based inkjet inks in particular are enjoying great popularity as colorants due to their environmental friendliness and the weather resistance of the pigments. Innovations in digital textile printing now enable a sustainable manufacturing process with a significant reduction in equipment, time, and labor costs when using pigment inks. Surface imaging is an opportunity to disrupt the current system of the textile printing industry and replace it with a more lucrative and holistic digital printing system in the future. With the development of nano inks, the invention of high-speed machines, one-step product development, and easy design flexibility, the future printing market will be digital printing for textiles [97,98].

The increased interest in AI (artificial intelligence) for inkjet printing technology is in manufacturing tasks such as localizing print patterns, creating high-resolution images for printing, and predicting print colors accurately [99].

Jasper and Ingle published a comprehensive study on transformative applications of deep learning and artificial intelligence in textile dyeing, printing, and finishing. Based on their study and similar research, it can be confirmed that the integration of deep learning and artificial intelligence into digital textile printing processes has progressed significantly. With the advancement of CNN-based algorithms, computer vision applications have become more adept at analyzing fabric images and aiding in quality control and pattern recognition. Genetic algorithms (GA) have been used to fine-tune neural networks to improve predictive capabilities. In addition, the emergence of generative adversarial networks (GANs) has opened up possibilities for generating creative patterns. However, amidst these advances, the lack of available data are a persistent challenge. The construction of robust deep learning models for textile processes is a multifaceted task, mainly due to the complex properties of textile materials. A particular challenge is the optimization of print quality to achieve batch uniformity and reduce errors. The integration of artificial intelligence can help to simplify the printing process. However, fine-tuning the model to account for variations in substrate materials and color properties is a constant challenge [100].

In the same area of applying AI and machine learning concepts in digital textile printing, Xiang et al. propose an improved AlexNet (the Convolutional Neural Network

(CNN) architecture known as AlexNet was developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton). The purpose was the automatic recognition of periodic primitive patterns from printed fabrics (smallest pattern units consisting of identical elements such as color, shape, and texture). The aim of the research was to simplify and facilitate the very labor-intensive and time-consuming manual process. The research carried out confirms the positive effect of the proposed method on the accuracy of feature extraction, although it requires a longer calculation time, which could limit its practical applicability in the textile industry. Future optimizations are essential to overcome this limitation and increase its feasibility [100,101].

The quality of color reproduction is also one of the problems in the application of digital printing technology to textiles. Li et al. are researching this very topic and propose two methods to improve the accuracy of color reproduction in digital textile printing: Full Process Optimization (FPO) and Neural Network Aided Original Adjustment (NNAOA). The FPO method optimizes the printing process by regulating ink consumption, i.e., by determining the least amount of ink that allows satisfactory reproduction without negative consequences such as color distortion. Another problem with digital textile printing is definitely dot gain (the problem of printed dots becoming larger than they should be). With textiles, this happens due to specific physical and mechanical properties such as porosity and capillarity. Not only do the printed dots become larger, but the shape of the dots is also deformed due to the influence of the structure and composition of the fabric. In the FPO method, dot gain is adjusted by adjusting the print settings to ensure that the final colors match the desired result. In the NNAOA method, neural network modeling is used to adjust the color values based on the measured Lab* values of the printed samples and the original CMYK values. The application of these methods has been shown to improve the accuracy of color reproduction in digital textile printing. However, a possible limitation could be that they depend on the precise calibration and optimization of printing parameters, which can require a lot of time and resources for effective implementation [102].

The detection of fabric defects in digital printing is also one of the most important tasks to ensure optimum final print quality. In general, eight common fabric defects are defined: traces of the passage of the print head, uncontrolled leakage of ink, gaps in the print, lint, water spots, shadows on the print, so-called ghosting, and streaks in the pattern (nozzle marks). In recent years, the rapid development of artificial intelligence has meant that manual recognition can be replaced by network models with deep learning algorithms. Models based on deep learning algorithms have been extensively studied and relevant research results can be found in the literature [100,103–106].

However, one of the limitations of the application is the lack of available datasets for the detection of fabric defects in digital printing. It can be said that there is still room for research in this area. Furthermore, two major challenges for the application of these models have been identified in previous research. First, real production requires the models to improve their detection accuracy as much as possible while ensuring real-time detection; and secondly, the goals of classification and localization are interrelated, so a link between classification and localization should be established for an excellent object detection model [100,107,108].

Some of the research on improved concepts for deep learning algorithms include that of Wu et al. who propose the application of the CARL (Classification-Aware Regression Loss)-YOLOF (You Only Look One-level Feature) model for detecting fabric defects in digital printing. In their study, they integrated the CARL concept into the YOLOF architecture to solve the problems of existing models and achieve higher detection accuracy while maintaining real-time applicability [100,104,109,110].

Su et al. proposed a high-quality model for fabric defect detection in digital printing, which they called FocusDet, in which they combined the faster-RCNN framework with innovative modules, namely Balanced Group Softmax (BAGS) and importance-based Sample Reweighting (ISR). The study addressed the challenges of class-imbalanced datasets, especially in the context of digital print tissue features with long-tailed distribution attributes. The same group of authors (Su et al.) proposed a model for detecting defects in digital printing fabrics based on the YOLOX architecture and integrating the squeeze-and-excitation (SE) attention module to improve real-time accuracy [111,112].

Hu et al. researched on apple object detection and localization based on improved YOLOX and RGB-D images, incorporating deep learning as a subset of machine vision. YOLOX is a state-of-the-art object detection framework based on deep learning, so the emphasis shifts toward deep learning within the context of machine vision. RGB-D images include both color (RGB) and depth (D) information, making them highly valuable for 3D object recognition and spatial understanding. Depth data comes from sensors like LiDAR or stereo cameras and can significantly enhance detection and localization accuracy. Using deep learning allows the system to learn from large datasets, adapting to variations in size, color, occlusion, and lighting conditions. These methods presented in the research of Hu et al. can be directly translated into textile applications, including defect detection, automated quality control, and optimization of resource use [24].

There is no doubt that digital technology has revolutionized fabric printing by enabling direct, precise, and sustainable designs on various textiles. This method has gained significant traction due to its environmental benefits, production efficiency, and ability to produce intricate, customized designs. Digital printing is undoubtedly reshaping the textile industry. It is precisely because of its great advantages that digital printing is predicted to grow significantly despite numerous obstacles and challenges. The benefits, including environmental friendliness, design flexibility, and high quality, are prompting many companies to adopt this technology as a forward-looking solution. For manufacturers seeking adaptability and sustainability, digital printing is an important step towards innovative, environmentally conscious production.

4. The Concepts of Sustainable Fashion and Apparel

The importance of the conceptual approach to sustainable production is no longer in question; sustainability is imperative, it is the paradigm of the future of the textile industry. Until recently, however, the focus of the textile industry has been on technological and cost aspects, lowering the price of the end product and increasing production efficiency. It is precisely this way of conducting business and consumers' habits of buying products easily and throwing them away quickly that have left the fashion and textile industry with a pollution problem: the textile industry is responsible for 20 percent of the world's pollution and ranks second on the scale of global polluters (behind oil refining and petroleum derivatives). Sustainable fashion is one measure to address this problem and is one of the most actively studied and utilized design methods; the 21st century fashion industry has shifted its paradigm towards sustainability, taking into account environmental impact and social and ethical responsibility [113,114].

As Geum and Park stated back in 2011 in their research and proposal for a new and systematic path to sustainable product-service integration, the need for sustainable industrialization must target sustainable design and production in order to achieve sustainability [114].

It is not so easy to define the meaning of the concept of sustainable production in a nutshell. The concept of SP goes beyond the definition and evaluation of environmental, economic, and social objectives for processes and products; it also encompasses the interaction between production systems [115,116].

The basis of the concept of sustainable production, which could be defined as the link between environmental, social, and economic frameworks, encompasses three equally important areas: ecology, economy, and society [86]. In any case, the concept of sustainable production is constantly evolving and changing contexts as new paradigms emerge. Sustainable production encompasses three pillars of sustainability: economy, society, and environment. Under the environmental aspect, sustainable production refers to the sustainable use of resources (raw materials, land, water, air, soil, landscape, and biodiversity) and energy sources, the promotion of a climate-neutral economy (reduction in greenhouse gas emissions), and the reduction in waste. The most important aspect of sustainable production is the sustainable use of resources and energy sources. Satisfying today's material needs should not lead to excessive extraction and degradation of these resources. Long-term sustainable use of resources requires taking into account their availability, ensuring the security of the supply chain and protecting ecosystems. At the same time, the environment's ability to absorb emissions and pollution must be maintained [117–119].

Advances in the field of sustainability have increased interest in the concept of sustainable design. This concept refers to eco-efficiency, reproduction, recycling, resource recovery, waste reduction, and life cycle assessment. Social design and eco-design are considered to be part of sustainable design. In the context of sustainable design, sustainable fashion can be defined as a concept that focuses on the design, production, and marketing of socially and environmentally responsible fashion products with the aim of reducing the negative impact of this industry on greenhouse gas emissions and climate change as well as social impact. The concept of sustainable fashion assumes that consumers develop an awareness of sustainable shopping and that the marketing concept is based on the observation and understanding of consumer behavior, purchasing patterns, and the discrepancy between attitudes and behavior. It involves the integration of sustainable materials and practices throughout the fashion supply chain. The goal of sustainable fashion is to achieve continuous product value by selecting appropriate fashion design elements that maintain product value and promote sustainability [120–124].

The principles of sustainable fashion include environmentally friendly production and distribution methods. The use of sustainable materials is also one of the most important aspects of sustainable fashion. This approach helps to raise awareness of the fashion industry's impact on the environment and promote more responsible practices. In the finishing processes, the focus is on the use of natural dyes, while in the construction of fabrics, the emphasis is on innovative materials. There is no doubt that the fashion industry is increasingly recognizing the need for sustainable change and is exploring alternative materials to achieve this [125].

The luxury fashion industry is placing increasing emphasis on sustainability because it has recognized the urgency of addressing the industry's environmental impact and because consumers are increasingly concerned. In addition to the positive impact of sustainability on consumers' perceived brand image, it also reflects a shift in consumer values. This makes it more effective for luxury brands than for fast fashion brands. Sustainability is also increasingly associated with luxury due to its higher cost and perceived intrinsic value. Therefore, luxury fashion brands are now aligning themselves with environmentally conscious practices and adopting sustainable product development practices such as exploring innovative designs and materials, including organic materials, natural dyes, bio-based materials, recycled textiles, and synthetic fabrics. These innovations reduce the industry's dependence on scarce resources and minimize its environmental footprint. In addition, brands are introducing cleaner and more efficient production processes, including techniques for wastewater disposal, greenhouse gas emission control, energy-saving methods, and reduced use of chemicals. Crucial to these sustainability efforts are factors

such as the durability and longevity of products and the adoption of gender-neutral and timeless designs, all of which aim to minimize consumption and reduce waste [126,127].

The digital transformation and the transition to a circular economy are among the basic prerequisites for adopting the concept of sustainable fashion. The circular economy approach to fashion aims to develop a more sustainable circular system where the goal is to extend the life of garments and preserve the value of products and materials for as long as possible. This means that all materials are recycled in several rounds. The products are designed to fit into a system where all aspects support the circular idea. The original design must take several life cycles into account. Materials must flow within the system and waste must be collected and valued as valuable material for recycling and recovery. All products must be taken back at the end of their useful life.

Digital support for the circular economy is certainly a concept of Industry 4.0, which has been proposing the concept of combining information and communication technology in production with digital technologies since 2011. The central idea of Industry 4.0 is to use new technologies to integrate all industrial processes into a flexible, efficient, and intelligent production process while ensuring high quality and lower costs. As mentioned earlier in this article, the leading technologies of the Industry 4.0 concept are additive manufacturing (AM), Cyber-Physical Systems (CPS), blockchain, artificial intelligence, artificial vision, big data and advanced analytics, cybersecurity, Internet of Things, robotics, and virtual and augmented reality. It is precisely these technologies that support the circular economy as they help to reduce material and energy consumption as well as waste and emissions generation. Acerbi and Taisch (2020) even conclude in their study that of all emerging technologies, additive manufacturing (AM) supports the concept of Industry 4.0 the most [127,128]. AM enables the recycling of a wider range of materials such as plastic, metal, and organic waste. Moreover, through the implementation of additive manufacturing, the design of circular products is carried out to facilitate the circularity of resources at the end of product life. In their study, they conclude that additive manufacturing is more energy and cost efficient compared to conventional production and AM is becoming one of the circular economy strategies. Digital technologies are also key to implementing other circular economy strategies such as resource recovery, recycling, and waste reduction [129–132].

In their research, Lannelongue et al. propose a simple and universally applicable method and tool to assess carbon footprints, which are also one of the focal points for the fashion industry's transition to a sustainable approach. They develop a freely available online tool, Green Algorithms, which allows users to estimate and report the carbon footprint of their calculations [132].

Ghobakhloo and Fathi (2021) argued that the digitalization of the energy sector, the digitalization of the manufacturing industry, and the introduction of new, smarter, and more sustainable products are the main opportunities for achieving a sustainable energy supply (which offers the most effective and balanced combination of economic, social, and environmental impacts) [133].

The application of business strategies guided by the principles of sustainable production, as well as the application of circular and sustainable design and waste management strategies, contribute to the further achievement of sustainable management and efficient use of natural resources by manufacturing companies; they reduce the release of pollutants, especially chemicals, and reduce waste.

5. Hypothesis Testing

In defining the concept of this theoretical review, the main research questions were set and hypotheses were formulated based on the preliminary phase in which the basic

supporting facts were defined. The hypotheses summarize the objective of this research by establishing a hierarchy of fundamental paradigms for the digital and sustainable transformation of the textile industry. Since the hypotheses are presented in the introductory section because they are chronologically set at the beginning of the research, it is possible to extract supporting facts as well as to confirm the hypotheses after the research has been completed.

- Facts supporting Hypothesis 1 can be summarized as follows:
 - The textile industry is transitioning from conventional manufacturing to smart, interconnected production systems, allowing for automation, real-time monitoring, and predictive maintenance.
 - Smart textiles, 3D printing, and digital simulation technologies improve material efficiency, reducing fabric waste and water consumption.
 - Digital textile printing is a key area of transformation, allowing for on-demand production, customization, and reduced resource consumption compared to traditional printing methods.
 - The introduction of e-commerce and virtual product presentation platforms has also accelerated digitalization in the apparel industry, reshaping business models and consumer interactions.
- Facts supporting Hypothesis 2 can be summarized as follows:
 - The core Industry 4.0 technologies that support sustainability in fashion and textile industries include additive manufacturing (AM), Cyber-Physical Systems (CPS), blockchain, artificial intelligence (AI), big data and advanced analytics, cybersecurity, Internet of Things (IoT), robotics, and virtual and augmented reality.
 - These technologies facilitate energy efficiency, resource optimization, and waste reduction, aligning industrial processes with circular economy principles.
 - There are significant resource savings and waste reductions through digital textile printing, IoT, and AI.
- Facts supporting Hypothesis 3 can be summarized as follows:
 - Regulations addressing hazardous substances, recycling mandates, and extended producer responsibility (EPR) are essential to ensure sustainability but require enforcement and adaptation across industries.
 - The transition to digital textile printing, AI, and smart manufacturing requires substantial investment in R&D, making collaboration with research institutions crucial.
 - Academia-industry partnerships help drive innovation, improve workforce training, and develop cost-effective solutions for digital sustainability.

As this is a review paper, which is a type of theoretical research, the hypotheses put forward cannot be tested quantitatively. However, based on the grouping of the supporting facts and the specification of the answers to the research questions, a qualitative validation of the hypotheses was carried out through a systematic analysis of the arguments and evidence presented in the text of the paper.

- In a scope of validation performed for Hypothesis 1, the document highlighted the use of CAD/CAM systems to conduct iterative testing in virtual models before physical production. This methodology reduced waste, errors, and unnecessary costs, allowing for faster production cycles and improved precision. The findings indicated significant time savings with CAD/CAM and waste reduction when using automated system. In addition, the document acknowledged that digital textile printing eliminated the need for traditional water-intensive dyeing processes, reducing water pollution and

chemical waste. Additive manufacturing (AM), which allows material recycling and an efficient use of resources reinforces the transition to circular economy principles.

- In a scope of validation performed for Hypothesis 2, the research highlighted the importance of integrating digital tools (such as IoT, AI, blockchain, and big data analytics) into traditional textile manufacturing to promote sustainability. These technologies enable traceability, ethical sourcing, waste reduction, and energy optimization. The document acknowledged that the adoption of Industry 4.0 in the textile industry faces significant challenges despite its benefits. High initial costs for smart manufacturing equipment and AI-driven production technologies make full-scale implementation difficult. Integration challenges exist, as traditional textile factories are often not equipped to incorporate IoT, automation, or AI-driven decision-making without substantial modifications. The hypothesis is strongly supported. While Industry 4.0 offers clear benefits, infrastructural, financial, and workforce-related barriers hinder its full-scale implementation in the textile sector.
- In a scope of validation performed for Hypothesis 3, the document emphasize interdisciplinary collaboration as a key factor in overcoming sustainability challenges in the textile industry. Research institutions drive technological advancements in sustainable textiles, including AI-driven defect detection, digital textile printing advancements, and new material development. Investment in education and training is necessary for workforce adaptation, requiring academic institutions and industry leaders to collaborate on skill development programs. The document confirms that collaboration among policymakers, industries, and academia is essential to successfully transition the textile industry toward sustainability.

6. Conclusions

By choosing environmentally friendly materials, improving energy efficiency, reducing waste, and promoting ethical practices, textile manufacturers can contribute to a more sustainable and responsible industry that meets the demands of today's conscious consumers and ensures a greener future for generations to come. Efforts to develop eco-design also increase the chances of reusing and recycling products, giving them a second life. Raising the level of digital knowledge and skills is extremely important for all players involved in the fashion and textile industries to improve business models and apply digital technologies in the field of fashion brand development but also in all segments of the supply chain. Innovations, interdisciplinary research, and digital technologies developed to meet the needs of the fashion industry are increasingly based on artificial intelligence. This opens up the possibility of extending the primary activity related to the traditional processes of manufacturing and selling fashion products to the field of digital product development for the needs of new generations of buyers and users of social networks and platforms and other creative industries.

Our conclusion can be summarized by the strategy, i.e., the rules, adopted by Viles et al. 2021 [134]. They are also cited by Zwiech, P. in her study and refer to the implementation of the concept of sustainable design and production based on the circular economy and the implementation of Industry 4.0: design for the circular economy; conservation of resources and preservation of their value; sustainable waste management; risk-free pursuit of the environment; prioritization of employee well-being; strengthening of management commitment to sustainability; positive contribution to the community; promotion of collaboration among value chain stakeholders; measurement and optimization of sustainable processes; promotion of the use of sustainable technologies [134,135].

Although there is still much room for progress, the timely recognition of the possibilities and opportunities that digital transformation can bring to fashion brands and all other

stakeholders involved in the fashion industry certainly already represents great potential for the development of future sustainable ecosystems for the production and distribution of clothing.

Digital transformation is key to the future of the textile and apparel industries, especially in the context of sustainability and responding to the demands of a modern, conscious market. However, based on a literature review and an examination of the results of various studies and analyses, it can be concluded that it is easier to analyze the theoretical framework of digital transformation than to actually apply and implement it. Due to a number of practical obstacles, especially in the textile and apparel industries, the majority of digital transformation projects are reduced to incremental improvements to existing technologies and processes, which is why there is no significant increase in value. Digital transformation is growing at a rate of more than 23% globally, and 89% of executives have made plans for a “digital first” approach to digital transformation. It is known that the key benefits of digital transformation are improved operational efficiency (40%), faster time to market (36%), and the ability to meet customer expectations (35%) on a global scale. However, the biggest challenge remains the lack of a digital strategy and the tendency to apply partial solutions. Investment in education and training is necessary to master new technologies, processes, and methods. The key to this is the use of an advanced digital platform that enables extended collaboration: with engineers, different departments, but also with customers, who today want to ‘feel’ the product virtually before buying it. Digital platforms must integrate tools for project management, 3D design, construction, simulation, and production planning. In this way, they effectively become a central place in manufacturing companies and ensure a single source of truth for all important information in the company, i.e., digital continuity. Recently, it has become increasingly apparent that a lack of knowledge and skills among employees is the biggest obstacle to successful digital transformation, which is why building the necessary capacity must be included.

This paper summarizes the concepts of the trends that are a necessary condition for the success of the textile and apparel industries—the application of the concepts of “Industry 4.0 (5.0)” and the “circular economy”. Both concepts require very complex interventions and their implementation is extremely complicated for any industrial company. Changes in the production process in line with these trends are primarily forced by the market and the technological development achieved, as well as by the EU through the provision of guidelines and framework deadlines for their implementation. The successful cooperation between educational institutions, institutes, companies, and other organizations for different types of production and finding successful economic application models are important in solving the often complex production requirements according to the EU directives. In addition to industry as the main actor in this transformation of the economy and production, the involvement of the government, regional authorities, experts, and the general public is required.

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